# ASSIGNMENT:02&03

# Artificial Intelligence

# Deadline: 18th April 2024

## Instructions:

1. It is group assignment of two members.
2. The assignment should be implemented using the Python programming language.
3. Strict adherence to academic integrity is required, with no plagiarism permitted. Any form of copying, whether from other students or online sources, will result in zero marks for the entire assignment.
4. The code must be well-commented for clarity and understanding.
5. Submissions must be made on time, as late submissions will be penalized.
6. Incomplete submissions lacking either the source code or the accompanying report will not receive any marks. The report should provide a concise explanation of the steps involved in the program for each question, along with the obtained results.

To submit the assignment, compress both the source code and the report into a single folder, named as ROLLNO_NAME.ZIP

**Question 1**

***Overview:***

You need to implement the Genetic Algorithm on dataset **UNSW-NB15** for the classification of intrusions**.**

**UNSW-NB15** is a network intrusion dataset. It contains nine different attacks, includes DoS, worms, Backdoors, and Fuzzers. The dataset contains raw network packets. The number of records in the training set is 175,341 records and the testing set is 82,332 records from the different types, attack and normal.

***Base Paper:***

Genetic Algorithm implementation details will be found from the paper:

**"An Evolutionary Approach to Intrusion Detection System using Genetic Algorithm"**

you must need to implement this paper on above mentioned Dataset by understanding the features of dataset and make changings into code. Moreover, you will get the better understanding regarding this question by reading this paper. Base Research Paper PDF

***Base code:***

The official implementation of above mentioned paper is here:

Git hub Repo

***Dataset:***

Data

***Requirements:***

1. At least three experiments must be done for evaluation by changings the parameters such as population size, crossover type and similar things related to genetic algorithm which could give useful insight.
2. A zip folder containing all the files of code with the clear and simple python run file which will initiate the code.
3. A precise report which will contain description of evaluations in tabular format, Evaluation Parameter should be same as base paper and your insightful findings. Describe those choices you could make while implementation which must be **relevant** just like which fitness function you are using and why.

**Confusions you might have in your mind.**

1. Do you really need to implement the same fitness function mentioned in base paper?

   NO, you can use any relevant fitness function.

2. Can't you make any changes in the base code means have to implement as it is?

   No, you can make any required changes which you feel is required for implementation of algorithm on required dataset but must be relevant and mentioned in report.

**Question 2**

You need to implement Multi-Variant Linear Regression model on self-created dataset of your 20 days' routine.

Dataset will contain multiple independent variables and one target/dependent variable

***Independent Variables***: *Sleep hours, Rank of food quality (based on heathy food), Working hours, Rank of Health Condition, Rank of mood (based on any outside tension).*

***Target/Dependent Variable:*** *Rank of Productivity.*

***Requirements:***

1. Zip folder containing data file and code file.
2. Implement multi-variant linear regression with Gradient descent from scratch. No library for the model.
3. Fill the values of data between 1 to 10.
4. Initialize ***m***, ***c and no_of_epochs*** randomly.
5. Perform the EDA and interactive data visualization steps, make the scatter plot for original data, Find the correlation of independent variables with target variables, plot the accuracies, plot the best fit line at each epoch, plot the working of gradient descent and plot the graph in between epoch and loss.